Battery-Constrained Maze Runner   Group #6

HARDWARE UNIT TESTS REPORT -

	Hardware name	Test results	Test code filename in git repo – UNIT TESTS sub folder	Referecnce for test code
1	M5StickC Plus + RoverC Base – 4-wheel mecanum drive	All 6 motion directions confirmed: FORWARD, BACKWARD, STRAFE R, STRAFE L, ROTATE CW, ROTATE CCW. Speed = 50/100 (setSpeed range: -100..100 per axis). Each motion ran 1500 ms then stopped 700 ms before next step. Direction labels and axis values printed to Serial @ 115200 baud and shown on M5 LCD.	01_RoverC_motors_test/01_RoverC_motors_test.ino	M5-RoverC library by M5Stack (github.com/m5stack/M5-RoverC). roverc.setSpeed(x, y, z): x = forward/back thrust, y = lateral strafe, z = yaw rotation; values – (-100)..(100)
2	MPU6886 IMU / Gyro (built into M5StickC Plus)	Gyro-stabilized straight drive: BASE_SPEED = 45/100 (nominal PWM for straight line); Kp = 0.8 (proportional gain scaling heading error to steering correction);	03_IMU_gyro_test/03_IMU_gyro_with_app_test.ino	M5.IMU (MPU6886) via M5StickCPlus library (github.com/m5stack/M5StickC-Plus). M5.IMU.SetGyroFsr(MPU6886::GFS_250DPS). Gyro integration (from updateYaw() in test): yaw += -(gyroZ - offset) * dt, dt clamped to [0, 0.1 s], wrapped to 0–360 deg. All integration constants and math were taken directly from the test code in the repo.
3	OPEN_SMART Serial MP3 Player (UART2: GPIO16 RX, GPIO17 TX, 9600 baud)	Module reset confirmed (UART2 response byte received within 50 ms). SD card selected via select_SD_CMD before any playback.	04_MP3_test_minimal/04_MP3_test_minimal.ino	OPEN_SMART MP3 Player A v1.0 serial command manual: static1.squarespace.com/static/584d41b3f5e2310b396cd953/t/5c7c2f29104c7b336a2f8380/1551642412037

		Volume tested at 3 levels: 2/30 (quiet), 15/30 (mid), 25/30 (loud). 3 audio files played sequentially by folder+index: play_filename(1,1), play_filename(1,2), play_filename(1,3). All played with clear audible output. 3 s playback window per file.		/Serial+MP3+Player+A+v1.0+Manual.pdf . Also: randomnerdtutorials.com/esp32-yx5300-yx6300-mp3-player-arduino/
4	WS2812B Addressable RGB LED Ring – Adafruit NeoPixel, GPIO12, 3 pixels	All 3 NeoPixels lit sequentially, one per loop iteration, in green (R=0, G=150, B=0). 500 ms delay between each pixel activation. Continuous loop confirmed. Protocol: NEO_GRB + NEO_KHZ800 (800 kHz single-wire). pixels.show() pushes updated state to ring after each setPixelColor call. All pixels respond correctly with no flickering.	05_LED_test/05_LED_test.ino	Adafruit NeoPixel library (github.com/adafruit/Adafruit_NeoPixel). Based on simple.ino example: https://github.com/adafruit/Adafruit_NeoPixel/blob/master/examples/simple/simple.ino Technion reference: https://github.com/ICST-Technion/IOT-Examples/tree/main/Neopixel_tutorial
5	3x VL53L1X Laser ToF Sensors (ESP32 devkit V1; SDA=GPIO21, SCL=GPIO22; XSHUT_1=GPIO5, XSHUT_2=GPIO23, XSHUT_3=GPIO18)	All three sensors initialized sequentially via XSHUT pins: sensor1 brought up first and re-addressed to 0x30, sensor2 initialized at 0x31, and sensor3 initialized at 0x32. No I2C address conflict. Continuous ranging started at 50 ms/sample (20 Hz). Distances from all three sensors (Left, Front, Right) printed over Serial @ 115200 baud in real time. Tested range: ~20 mm – ~2000 mm, all three sensors reading simultaneously and independently.	three_sensors_working/three_sensors_working.ino	VL53L1X library by Pololu (github.com/pololu/vl53l1x-arduino). Technion IOT Examples reference: https://github.com/ICST-Technion/IOT-Examples/tree/main/hardware%20unit%20tests/VL53L1X